

Solution Requirements

Date	20 July 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Data Input	Users can enter N, P, K, temperature, humidity, pH, and rainfall values via a web form	High
2	Data Validation	System validates that all inputs are numeric and within realistic agronomic ranges	High
3	Model Inference	System runs the trained ML model (KNN / Logistic Regression) on validated input and returns a crop recommendation	High
4	Clustering Insight	System uses K-Means clustering to group similar soil/climate profiles for exploratory analysis	Medium
5	Result Display	System displays the recommended crop clearly on a results page/section	High
6	Model Persistence	The best-performing trained model is saved (pickle) and reloaded at runtime rather than retrained per request	High
7	Reporting	System can optionally show confidence/probability of the prediction	Medium
8	Other	Application should be usable without login (single-session, stateless prediction)	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Prediction should return quickly after form submission	Response in < 2 seconds
2	Scalability	System should handle multiple concurrent users during demo/testing	Supports 50+ concurrent requests
3	Security & Data Privacy	No personal farmer data is stored beyond the session	No persistent storage of user-submitted input
4	Reliability & Availability	Application should run consistently without crashing on invalid input	Graceful error handling, 0 unhandled exceptions in testing
5	Usability & Accessibility	Simple, responsive form usable on desktop and mobile browsers Code should be modular	Mobile-responsive layout, clear labels/tooltips
6	Other (Maintainability)	(separate preprocessing, model, and app layers) for	Clear folder separation as per repo structure

